Mitali Joshi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.14
The Vidhyanjali International School	Class XII (CBSE)	2020	84.80%
The Vidhyanjali International School	Class X (CBSE)	2018	73.60%

PROJECTS

M.Tech Research | DRDO-DRDL Funded Project

Study the effect of unsteady pressure fluctuation in sub-cavity for Mach No. 2.5.

- Conducted high-speed aerodynamic investigations and passive flow control experiments in a supersonic cavity flow environment to mitigate dynamic pressure fluctuations and flow-induced acoustic tone generation.
- Evaluated **8 distinct passive control configurations** across multiple flow conditions, achieving a **68% to 89% reduction** in pressure standard deviation, mean pressure, and peak frequency power content.
- Developed custom MATLAB workflows for post-processing static pressure and acoustic signal data, establishing standardized data extraction and spectral analysis.
- Built MATLAB scripts for **Schlieren image processing**, short-time Fourier transforms (STFT spectrograms), and spatial-temporal correlation plotting to analyze shock wave movement and unsteady shear layer behaviors.
- **Transonic Flow Extension:** Extended post-processing data analysis frameworks to analyze unsteady transonic cavity flow dynamics, processing high-frequency sensor streams to characterize acoustic tone resonance across regimes.
- **Wind Tunnel Calibration & Test Model Engineering:** Designed and fabricated high-precision model, sub-millimeter accurate model prototypes using an **Anycubic Photon M7 Max (SLA resin)** for ultra-smooth surface finishes and an **FDM printer (Bambu Lab A1)** for rapid structural prototyping, integrating real-time **pressure transducers** directly into resin-printed internal housings for flush-surface acoustic measurement.
- Engineered sacrificial dummy test models to optimize structural integrity and sensor placement during initial tunnel calibration runs, protecting expensive dynamic instrumentation from high dynamic pressure loads during start-up shock.

B.Tech Thesis Project

Design and Analysis of CD Nozzle and Chevron Nozzle for Aeroacoustics

- Designed and optimized a **Mach 3 convergent-divergent (CD) nozzle** integrated with triangular chevron trailing-edge modifications aimed at suppressing high-speed jet noise emissions.
- Modeled CAD geometries in **CATIA V5** and conducted structured grid generation to capture fine mixing layers at the chevron interface.
- Performed 2D compressible CFD numerical simulations in **ANSYS Fluent** to analyze flow, demonstrating a **25% noise reduction** while minimizing thrust efficiency losses.

Determine Burning rate of Composite Solid Propellant | Propulsion Course

- Executed experimental characterization to measure **the linear burning rate of composite solid propellant** is a combination of oxidizer, binder, and metal additives across **varying chamber pressure conditions** using a high-pressure strand burner setup.
- Derived empirical burn rate coefficients and pressure exponents using Saint-Robert's (Vieille's) Law to evaluate performance.

- The propellants formulation contains 80% total solids loading and 20% binder system. Ammonium perchlorate (AP), serves as the oxidizer, while the binder is composed of hydroxyl terminated polybutadiene (HTPB) cured with toluene diisocyanate (TDI), plasticizer di-octyl adipate (DOA).

Numerical Solver Development | Advanced CFD Course

- Developed custom **MATLAB algorithms** using finite difference methods (FDM) to solve partial differential equations governing viscous incompressible fluid flow and unsteady thermal conduction.
- **Lid-Driven Cavity Flow:** Formulated stream function-vorticity equations using first-order and second-order upwind schemes for non-linear convection terms across Reynolds numbers (Re = 100, 400, 1000). Validated

horizontal/vertical velocity profiles against published benchmark data by Ghia et al., in 1982.
- **Heat Conduction Solvers:** Implemented an FTCS (Forward Time Centered Space) explicit scheme for 2D slab heat conduction (2L thickness) across Biot numbers and Fourier numbers, verifying numerical convergence against analytical Heisler charts.
- Generated contour plots, velocity vectors, and streamlines to visualize flow recirculation zones and dynamic thermal boundary layer evolution.

Flow study for different applications | Aerodynamics & Compressible Flow Course

- Executed numerical simulations of 2D planar unsteady compressible shock tube dynamics in **ANSYS Fluent**, validating wave speeds, and pressure jump ratios against theoretical 1D shock tube equations.
- Operated optical **Schlieren visualization systems** to record high-speed shock wave structures, expansion fans, and boundary layer separation over spiked bodies in hypersonic flow regimes.
- Analyzed unsteady vortex shedding (St), drag coefficient, and lift coefficient past a 2D circular cylinder to quantify dynamic wake structures.

Stability Analysis | Flight Mechanics Course

- Conducted comprehensive longitudinal, lateral, and dynamic stability evaluations for fixed-wing aircraft and glider configurations under varying aerodynamic loading.
- Evaluated performance envelopes by modeling thrust required vs. thrust available to compute optimum cruise altitude maintenance under engine RPM variations.
- Also compared and tested Dihedral and Anhedral Glider Configuration for range and stability.

Evaluated ML Algorithm performance | Machine Learning Applications Course

- Built, trained, and evaluated supervised machine learning algorithms (**Logistic Regression, ANNs, CNNs**) using **Python liberary** for mechanical property estimation and material design classification.
- Processed physical property datasets (stress-strain parameters, Young's modulus, elastic limit) across 1,000 samples to automate material suitability ratings for EV chassis construction, achieving **95% prediction accuracy**.
- Constructed deep Convolutional Neural Networks (CNNs) for visual image classification on benchmark engineering datasets (CIFAR-10), benchmarking performance using confusion matrices, precision, recall, and loss curves.

AREA OF EXPERTISE

Experimental Techniques: Supersonic Wind Tunnel Calibration, Pressure Measurement, Schlieren Imaging

Software: ANSYS Fluent, ANSYS CFD

Design/Modelling: CATIA V5, Fusion 360, AutoCAD, SolidWorks

Data Processing: MATLAB (Post-Processing & Numerical Solvers)

Domain Expertise: Aerodynamics, Aeroacoustics, Flow Analysis

PUBLICATIONS & CONFERENCES

- **Joshi, M.**, et al. (2022). Impact of Plastic Bullet for INSAS on a target with trajectory Correction. International Journal of All Research Education and Scientific Methods (IJARESM), 10(7), Impact Factor: 7.429, A.
- **Joshi, M.**, et al. Suppression of acoustic wave using passive device in supersonic cavity with sub-cavity. Accepted at **International Symposium on Shock Waves 35 (ISSW 35), Brisbane, Australia,** July 2025.
- **Joshi, M.,** et al. (2026). A Guide for Post-Processing ANSYS-Fluent Data in MATLAB. Zenodo. DOI
- **Joshi, M.,** et al. Suppression of acoustic wave using a passive technique in a supersonic cavity-sub-cavity flow. Poster presented at: **Aerospace Engineering Research and Educational Symposium (AERES '26)**, Indian Institute of Science (IISc), Bangalore, India, Dec 2025.
- **Joshi, M.**, et al. Investigation of effect of sub-cavity and acoustic attenuation. 9th **National Symposium on Shock Waves (NSSW), Jaipur**, Dec 2026.
- **Joshi, M.**, et al.Numerical Investigation of Unsteady Flow Dynamics over Double-Disk Aerospikes at Mach 2.0. 9th **National Symposium on Shock Waves (NSSW), Jaipur**, Dec 2026.
- **Joshi, M.,** et al. Experimental and Computational study for supersonic cavity flow over a deep sub-cavity and passive ventilation control. 6th **National Aerospace Propulsion Conference (NAPC),** IIT Hyderabad, Jan 2027.

HONORS AND AWARDS

MIRAI Setu Fellow (Japan-India Government Exchange):

Selected by AFS for a 1-month funded program (Oct-Nov 2026) focusing on Japanese mechanical industry standards, advanced technology exposure, and cultural exchange.

1st Place Winner - Technical Expo 2023:

Awarded top position at Parul University Tech Expo for designing and presenting an innovative engineering project.

Merit-Based Academic Scholarship:

Awarded a 40% tuition scholarship by Parul University in recognition of outstanding academic performance.

Most Innovative Idea Award - CBSE Bal Vigyan Maths (2020):

Received top honours at the CBSE Cluster level for Financial Mathematics research presentation.

INTERNSHIPS

AI Trainee and Evaluation | Soul AI | Freelancer

- Delivered detailed technical feedback and structured dataset annotations, enhancing model reliability and reducing error rates in automated key-moment detection systems.
- Standardized validation pipelines for high-precision data labelling, ensuring consistent quality control across streaming analytics tools.

International Admissions Counselor | Parul University, Vadodara

- Structured and managed a comprehensive admissions database for applicant and enrolled student records, ensuring **100% data accuracy** and seamless accessibility.
- Analysed **500+ student skill profiles** and academic interests to provide data-driven program recommendations, streamlining applicant onboarding.

POSITIONS OF RESPONSIBILITY

Placement Coordinator | Office of Career Services (OCS), IIT Hyderabad

- Managed campus placement drives and employer outreach for the Mechanical and Aerospace Department.

Teaching Assistant | Experimental and Computational Laboratory, IIT Hyderabad

- Mentored M.Tech students in Computational Fluid Dynamics (CFD) fundamentals and instructed post-processing routines in MATLAB for complex fluid flow data analysis.

Volunteer for Teaching | Rural Development Program by Telangana Government, IITH

- Instructed local school students in Communication Skills and Mathematics under the Telangana Government Rural Development Program, improving foundational literacy and problem-solving skills through structured lessons.

Deputy Coordinator & Volunteer | Placement & Mock Exams, IIT Hyderabad

- Organized mock technical exams and interview preparation sessions for Mechanical and Aerospace students.

Fest Coordinator | Dhoom (Cultural) & Projection (Technical), Parul University

- Spearheaded event scheduling, logistics, and student coordination for university-wide technical and cultural fests.

CERTIFICATES

- Ornithopter Fabrication and Flight Evaluation: Hands-on Training on Ornithopter Design.
- UAV Design & Development: Hands-on Workshop on Design and Development of Quadcopter UAV for Payload Delivery (Fusion 360 Structural Modeling & Dynamic Load Analysis).
- Military & Leadership: NCC 'C' Certificate Holder (Air Wing).

HOBBIES AND INTERESTS

Reading || Poetry || Innovation || Photography & Digital Media Creation || Strategical Board Games || Structural Organization & Decluttering || Journaling & Personal Development || Technical Content Writing || Storytelling || Teaching with fun.